

FMUSER FUQ-44105B 16-Channel DVB-C QAM Modulator with 16 DVB-C Tuners 512 IP Inputs 16 RF Outputs ETR289 Scrambling Multiplexing for CATV IPTV Headend



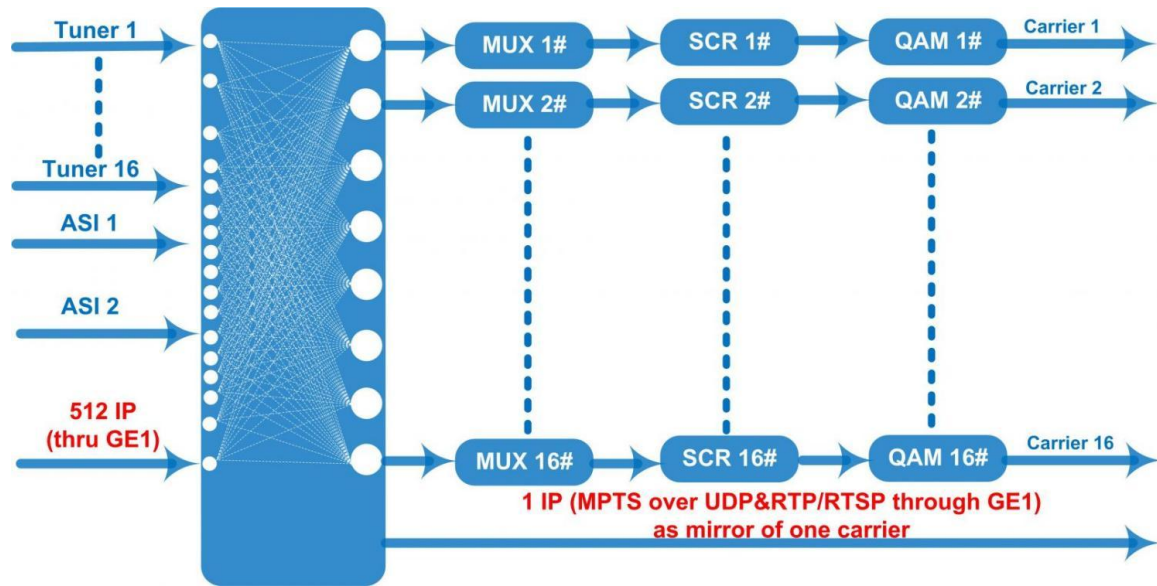
Product overview

The FMUSER FUQ-44105B is a high-density QAM modulator supporting 16x DVB-C tuner inputs, 2x ASI (BNC), and 512x IP streams via GE1 (UDP/RTP). It performs multiplexing, ETR289-compliant scrambling, and 16-256QAM modulation, delivering 16x RF outputs (50-960MHz) and 1x MPTS IP mirror (UDP/RTP/RTSP). The 1U rack-mount unit features MER ≥ 40 dB and web-based control. Ideal for CATV digitization, IPTV redistribution, and multi-standard headend deployments. FU-44105C is also characterized with high integrated level, high performance and low cost. It supports dual power supply (optional). This is very adaptable to newly generation CATV broadcasting system.

Key feature

Broad Compliance: Supports DVB-C/T/S/S2/S2X, ATSC, ISDB-T, and J.83A/B/C standards

Working Principle



Specifications

Input	16 DVB-C (DVB-T/-(T)/-S/-S2/-S2 X, ATSC, ISDB-T Optional) FTA Tuner	16 DVB-C (DVB-T/-(T)/-S/-S2/-S2 X, ATSC, ISDB-T Optional) FTA Tuner	16 DVB-C (DVB-T/-(T)/-S/-S2/-S2 X, ATSC, ISDB-T Optional) FTA Tuner	16 DVB-C (DVB-T/-(T)/-S/-S2/-S2 X, ATSC, ISDB-T Optional) FTA Tuner	16 DVB-C (DVB-T/-(T)/-S/-S2/-S2 X, ATSC, ISDB-T Optional) FTA Tuner	16 DVB-C (DVB-T/-(T)/-S/-S2/-S2 X, ATSC, ISDB-T Optional) FTA Tuner
Input	512 IP input through GE1 over UDP and RTP protocol	512 IP input through GE1 over UDP and RTP protocol	512 IP input through GE1 over UDP and RTP protocol	512 IP input through GE1 over UDP and RTP protocol	512 IP input through GE1 over UDP and RTP protocol	512 IP input through GE1 over UDP and RTP protocol
Input	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface
Tuner Section	Multi-mode tuners switchable (New)	DVB-C	Standard	Standard	J.83A(DVB-C), J.83B, J.83C	J.83A(DVB-C), J.83B, J.83C
Tuner Section	Multi-mode tuners switchable (New)	DVB-C	Frequency In	Frequency In	60~890MHz	60~890MHz

Tuner Section	Multi-mode tuners switchable (New)	DVB-C	Constellation	Constellation	16/32/64/128/256 QAM	16/32/64/128/256 QAM
Tuner Section	Multi-mode tuners switchable (New)	DVB-T/(T)	Frequency In	Frequency In	60~890MHz	60~890MHz
Tuner Section	Multi-mode tuners switchable (New)	DVB-T/(T)	Bandwidth	Bandwidth	6/7/8 M bandwidth	6/7/8 M bandwidth
Tuner Section	Multi-mode tuners switchable (New)	ISDB-T	Frequency In	Frequency In	60~890MHz	60~890MHz
Tuner Section	Version 1	DVB-S/S2	Frequency In	Frequency In	950~2150 MHz	950~2150 MHz
Tuner Section	Version 1	DVB-S/S2	Symbol rate	Symbol rate	QPSK 1~45 Mbauds 8PSK 1~45 Mbauds	QPSK 1~45 Mbauds 8PSK 1~45 Mbauds
Tuner Section	Version 1	DVB-S/S2	Code rate	Code rate	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	Version 1	DVB-S/S2	Constellation	Constellation	QPSK, 8PSK	QPSK, 8PSK
Tuner Section	Version 2 (New)	DVB-S	Frequency In	Frequency In	950~2150 MHz	950~2150 MHz
Tuner Section	Version 2 (New)	DVB-S	Symbol rate	Symbol rate	0.5~45Mps	0.5~45Mps
Tuner Section	Version 2 (New)	DVB-S	Signal Strength	Signal Strength	-65~-25dBm	-65~-25dBm
Tuner Section	Version 2 (New)	DVB-S	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	Version 2 (New)	DVB-S	Constellation	Constellation	QPSK	QPSK

Tuner Section	Version 2 (New)	DVB-S	Max input bitrate	Max input bitrate	≤129 Mbps	≤129 Mbps
Tuner Section	Version 2 (New)	DVB-S2	Frequency In	Frequency In	950~2150 MHz	950~2150 MHz
Tuner Section	Version 2 (New)	DVB-S2	Symbol rate	Symbol rate	QPSK/8PSK /16APSK: 0.5~45 Msp 32APSK: 0.5~40Msp;	QPSK/8PSK /16APSK: 0.5~45 Msp 32APSK: 0.5~40Msp;
Tuner Section	Version 2 (New)	DVB-S2	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	Version 2 (New)	DVB-S2	Constellation	Constellation	QPSK, 8PSK, 16APSK, 32APSK	QPSK, 8PSK, 16APSK, 32APSK
Tuner Section	Version 2 (New)	DVB-S2	Max input bitrate	Max input bitrate	≤129 Mbps	≤129 Mbps
Tuner Section	Version 2 (New)	DVB-S2X	Frequency In	Frequency In	950-2150MHz	950-2150MHz

Tuner Section	Version 2 (New)	DVB-S2X	Symbol rate	Symbol rate	QPSK/8PSK /16APSK: 0.5~45 Msps 8APSK K/32APSK: 0.5~40Mps	QPSK/8PSK /16APSK: 0.5~45 Msps 8APSK K/32APSK: 0.5~40Mps
Tuner Section	Version 2 (New)	DVB-S2X	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10, 2/3-L, 32/45, 11/15, 7/9

Tuner Section	Version 2 (New)	DVB-S2X	Constellation	Constellation	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK
Tuner Section	Version 2 (New)	DVB-S2X	Max input bitrate	Max input bitrate	≤129 Mbps	≤129 Mbps
Tuner Section		ATSC	Frequency In	Frequency In	30~1000M Hz	30~1000M Hz
Tuner Section		ATSC	Bandwidth	Bandwidth	6M	6M
Tuner Section		ATSC	Constellation	Constellation	8VSB	8VSB
Multiplexing	Maximum PID Remapping	Maximum PID Remapping	360 output per channel	360 output per channel	360 output per channel	360 output per channel
Multiplexing	Function	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Function	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically
Scrambling Parameters	Max simulcrypt CA	Max simulcrypt CA	4	4	4	4
Scrambling Parameters	Scramble Standard	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters	Connection	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection

Modulation	QAM Channel	QAM Channel	16 non-adjacent carriers output	16 non-adjacent carriers output	16 non-adjacent carriers output	16 non-adjacent carriers output
Modulation	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	MER	MER	≥40db	≥40db	≥40db	≥40db
Modulation	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	RF output level	RF output level	-20~+10dbm(87~107dbμV), 0.1db step	-20~+10dbm(87~107dbμV), 0.1db step	-20~+10dbm(87~107dbμV), 0.1db step	-20~+10dbm(87~107dbμV), 0.1db step
Modulation	Symbol Rate	Symbol Rate	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping
Modulation			J.83A	J.83A	J.83A	J.83B
Modulation	Constellation	Constellation	16/32/64/128/256QAM	16/32/64/128/256QAM	16/32/64/128/256QAM	64/256QAM
Modulation	Bandwidth	Bandwidth	8M	8M	8M	6M
Stream out	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)	16 RF output (F type interface)
Stream out	1 IP (MPTS) output over UDP and RTP/RTSP (GE1 only), as mirror of one carrier	1 IP (MPTS) output over UDP and RTP/RTSP (GE1 only), as mirror of one carrier	1 IP (MPTS) output over UDP and RTP/RTSP (GE1 only), as mirror of one carrier	1 IP (MPTS) output over UDP and RTP/RTSP (GE1 only), as mirror of one carrier	1 IP (MPTS) output over UDP and RTP/RTSP (GE1 only), as mirror of one carrier	1 IP (MPTS) output over UDP and RTP/RTSP (GE1 only), as mirror of one carrier
System	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)

System	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language
System	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
General	Dimension(W*D*H)	Dimension(W*D*H)	Dimension(W*D*H)	482mm×300mm×44.5mm	482mm×300mm×44.5mm	482mm×300mm×44.5mm
General	Temperature	Temperature	Temperature	0~45■(Operation) ; -20~80■(Storage)	0~45■(Operation) ; -20~80■(Storage)	0~45■(Operation) ; -20~80■(Storage)
General	Power	Power	Power	AC 100V±1050/60Hz; AC 220V±10%, 50/60HZ	AC 100V±1050/60Hz; AC 220V±10%, 50/60HZ	AC 100V±1050/60Hz; AC 220V±10%, 50/60HZ

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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